Meeting 3 SP23

Point Cloud Acceleration

@crawfish

Outline

Background

PointAcc

Evaluation

Section 1

Background

Machine Learning

Like terminator but significantly less cool

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- Incredibly applicable → relevant to just about anything

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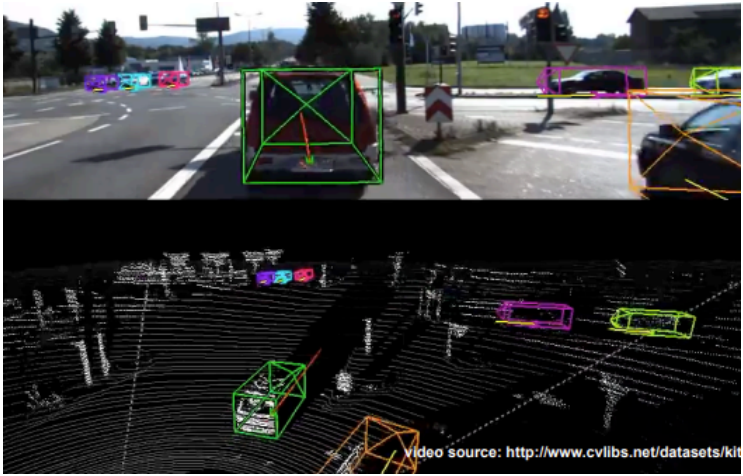
  - AR/VR

AV Driving



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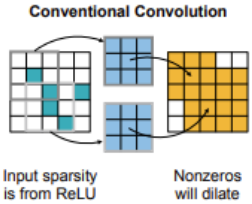
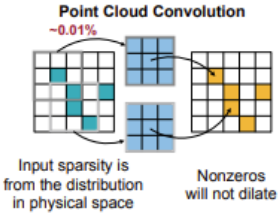
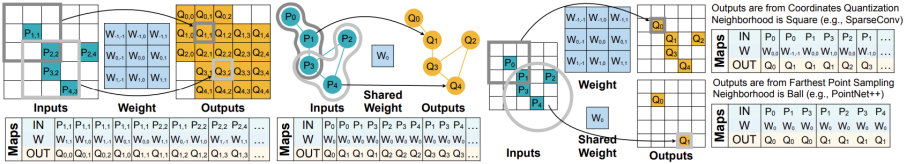




Point Cloud Convolution



(a) Conventional Convolution
(inputs **quickly become denser**)

(b) Graph Convolution
(maps are **given** and **fixed**)

(c) Point Cloud Convolution
(inputs **keep** sparse; maps are **explicitly calculated** during inference)

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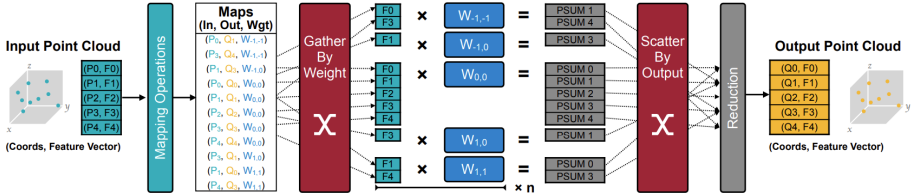
  - Must explicitly calculate this **every iteration** for point clouds, can change during down-sampling
- Data is irregular, unstructured, difficult to work with
- Much research has been done on improving the point cloud convolution, open problem

Point Cloud Convolution



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3. Large memory footprint

Section 2

PointAcc

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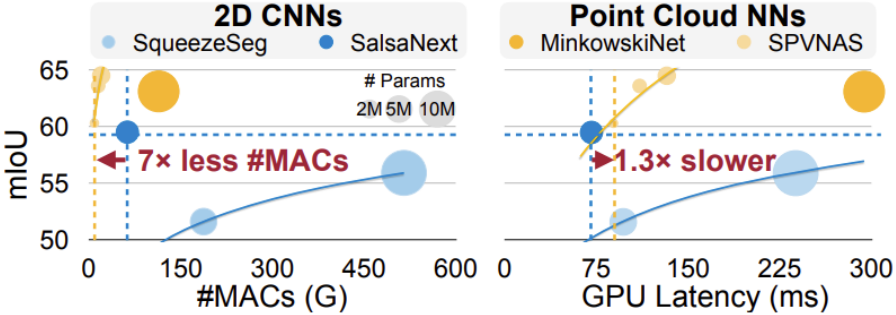
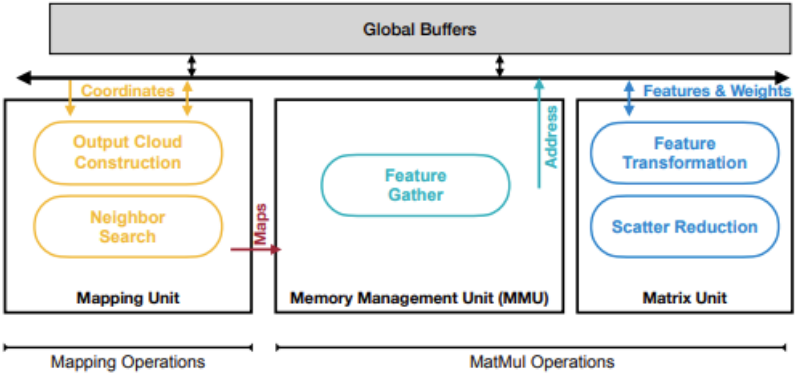
  - Point cloud networks cannot be run efficiently on current GPU designs
  - Latency of point clouds is higher on GPU than 2D CNNs $\rightarrow$ but 7x fewer MACs required to evaluate point cloud networks
  - Implies there must be inefficiency somewhere

PointAcc


2D CNNs
SqueezeSeg
SalsaNext
Params
2M 5M 10M
7× less #MACs
mIoU
65
60
55
50
0
150
300
450
600
#MACs (G)
Point Cloud NNs
MinkowskiNet
SPVNAS
1.3× slower
0
75
150
225
300
GPU Latency (ms)

PointAcc


Global Buffers
Coordinates
Features & Weights
Output Cloud Construction
Neighbor Search
Maps
Feature Gather
Address
Feature Transformation
Scatter Reduction
Mapping Unit
Memory Management Unit (MMU)
Matrix Unit
Mapping Operations
MatMul Operations

Mapping Unit

- **Purpose:** Generate a mapping between input point, output point, and weight index based on distance comparisons

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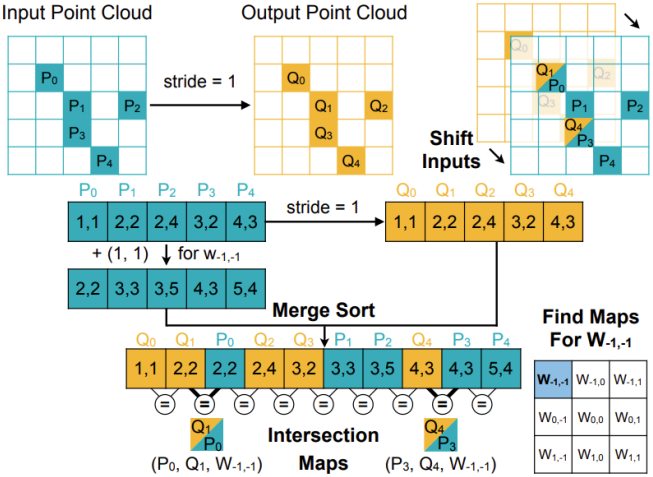
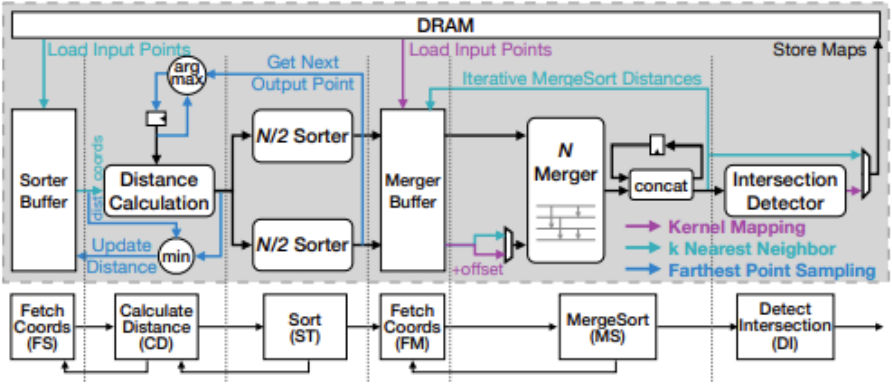
  - Low memory requirements
  - Easily parallelizable

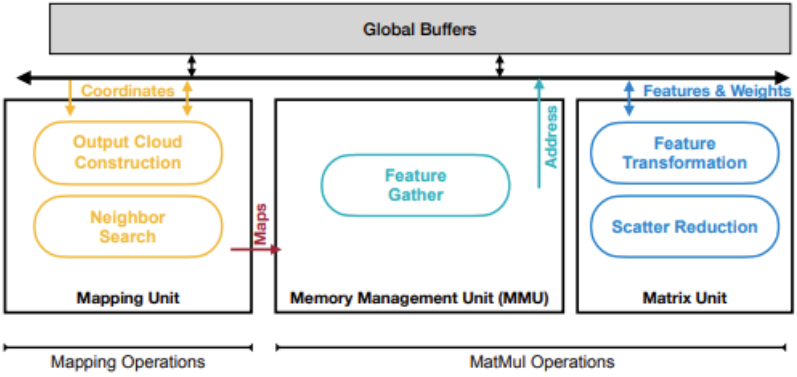
Mergesort


Input Point Cloud
Output Point Cloud
stride = 1
Shift Inputs
P0 P1 P2 P3 P4
1,1 2,2 2,4 3,2 4,3
stride = 1
Q0 Q1 Q2 Q3 Q4
1,1 2,2 2,4 3,2 4,3
+ (1, 1)
for w-1,-1
2,2 3,3 3,5 4,3 5,4
Merge Sort
Q0 Q1 P0 Q2 Q3 P1 P2 Q4 P3 P4
1,1 2,2 2,2 2,4 3,2 3,3 3,5 4,3 4,3 5,4
Intersection Maps
(P0, Q1, W-1,-1)
(P3, Q4, W-1,-1)
Find Maps For W-1,-1
W-1,-1 W-1,0 W-1,1
W0,-1 W0,0 W0,1
W1,-1 W1,0 W1,1

Mapping Unit Architecture



PointAcc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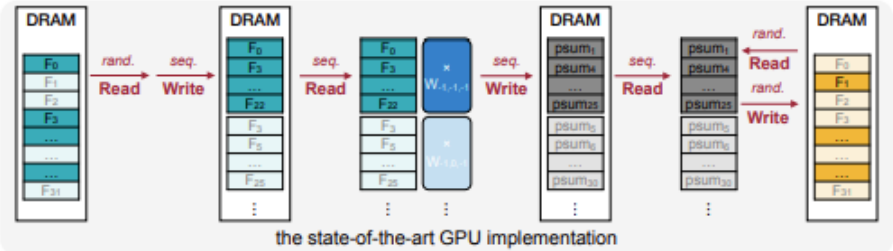
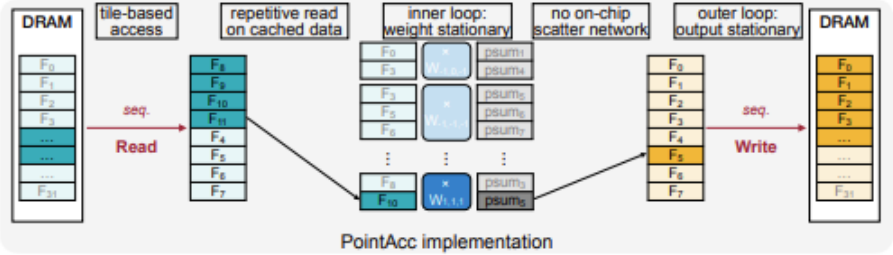
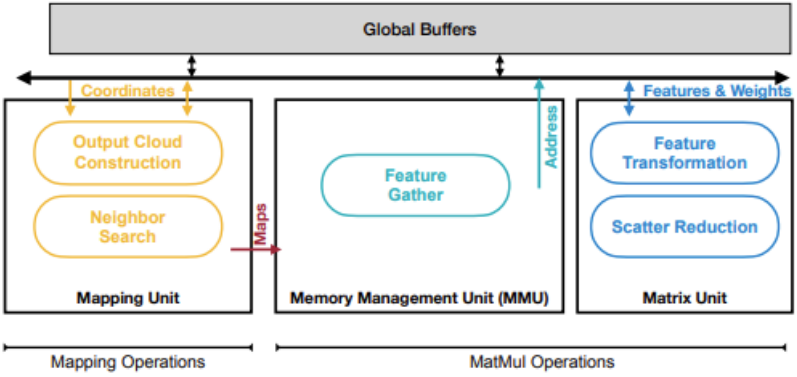
- Temporally fuse computation of consecutive FCs in point cloud convolution

MMU


DRAM
rand.
Read
seq.
Write
seq.
Read
seq.
Write
seq.
Read
rand.
Read
rand.
Write
the state-of-the-art GPU implementation


DRAM
tile-based access
repetitive read on cached data
inner loop: weight stationary
no on-chip scatter network
outer loop: output stationary
DRAM
seq.
Read
seq.
Write
PointAcc implementation

PointAcc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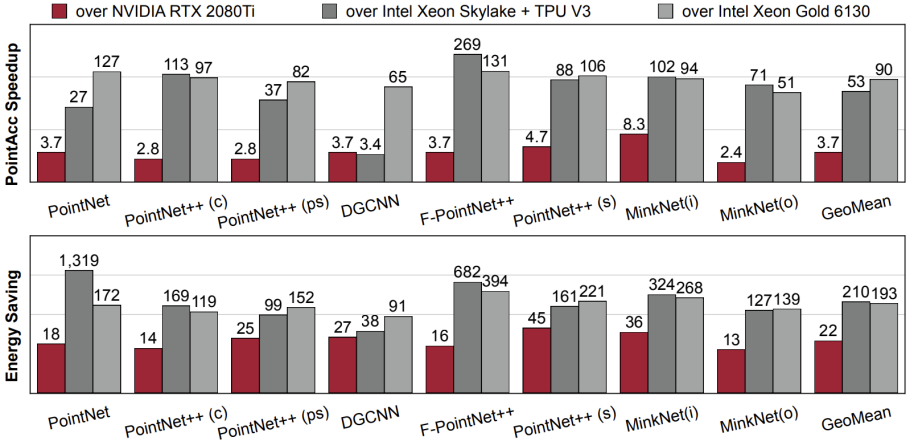
- Very similar to what we see in TPUs $\rightarrow$ not much new here

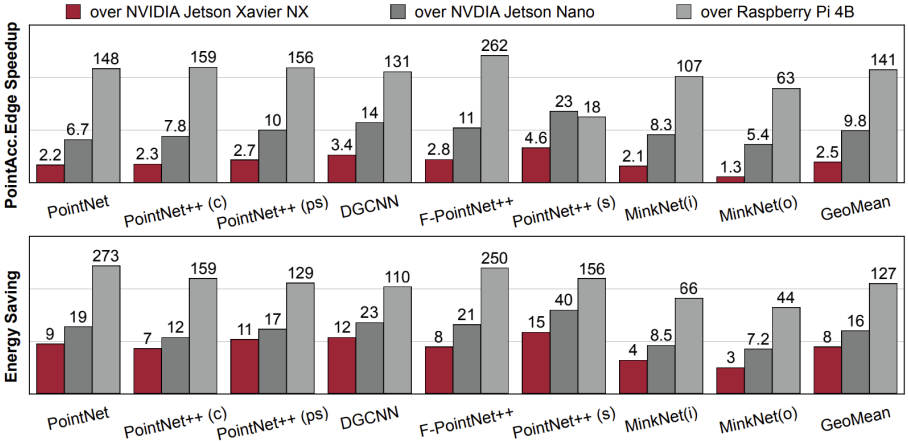
Section 3

Evaluation

Server Products



Edge Devices



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- Set of minor extensions to CNN accelerators intended to be integrated into SoC systems
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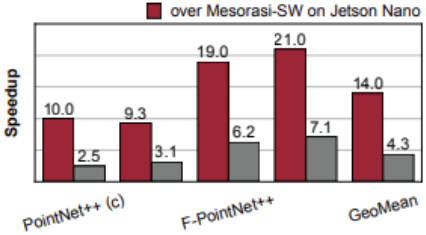
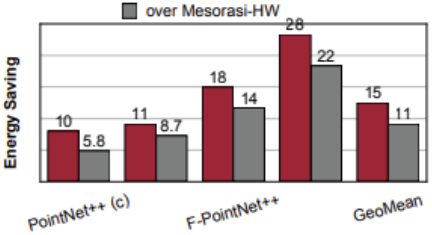
- PointAcc can provide 9.1% higher accuracy (mIoU), 130x lower latency

Mesorasi


over Mesorasi-SW on Jetson Nano
Speedup
10.0
2.5
9.3
3.1
19.0
6.2
21.0
7.1
14.0
4.3
PointNet++ (c)
F-PointNet++
GeoMean


over Mesorasi-HW
Energy Saving
10
5.8
11
8.7
18
14
28
22
15
11
PointNet++ (c)
F-PointNet++
GeoMean

The end.